CV

Satyaki Mukherjee

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Singapore, Singapore - 267745
Date of birth: 18-06-1992
Birth city: Durgapur, West Bengal, India
Highest Degree: Ph.D. in Mathematics
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| *Surname:* Mukherjee
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| *Nationality:* Indian
| *family status :* Unmarried

1 Current Affiliation

Postdoctorate @ National University of Singapore
Mentor: Subhroshekhar Ghosh

2 Past Affiliations

2.1 Employment

Postdoctorate @ Theoretical Foundations of AI group at Technical University of Munich — January 2022 - December 2023
Mentor: Debarghya Ghoshdastidar

2.2 Education

University of California, Berkeley
Ph.D.,Department of Mathematics — Aug 2015 - May 2021
Advisor: Professor Nikhil Srivastava
Thesis : Pseudospectra of matrices and Pointspectra of infinite operators
GPA : 3.8/4

Indian Statistical Institute, Bangalore, India
Master of Mathematics (Gold Medalist) — July 2014 - May 2015
Total Percentage: 87%
Bachelor of Mathematics — July 2011 - May 2013
Total percentage score: 90%

Programming Languages Known: Python, Octave, C++.

3 Current Projects

I am currently working on a range of estimation and hypothesis testing problems, including topics related to the nonparametric maximum likelihood estimator (NPMLE) and multiresolution analysis (MRA). In parallel, I am pursuing several machine learning projects—one focused on reducing model complexity through parameter sparsification, and another exploring the use of transformer architectures to convert news sentiment into structured data for stock market prediction.

4 Preprints

* starred contributors have equal contribution and are in alphabetical-order

1. Approximate FKG inequalities for phase-bound spin systems **Satyaki Mukherjee***, Vilas Winstein* Arxiv 2026

2. **G**aussian mixtures and non-parametric likelihoods through the lens of statistical mechanics Subhroshekhar Ghosh*, Adityanand Guntuboyina*, **Satyaki Mukherjee***, Hoang-Son Tran* Arxiv 2026

5 List of Published/Accepted papers

* starred contributors have equal contribution and are in alphabetical-order

1. **On a generalisation of the coupon collector problem** Siva Athreya*, **Satyaki Mukherjee***, Soumendu Sundar Mukherjee* JTP 2025 Arxiv 2023
2. **Recovering Imbalanced Clusters via Gradient-Based Projection Pursuit** Martin Eppert*, Debarghya Ghoshdastidar*, **Satyaki Mukherjee*** Arxiv 2025 (*Accepted at **JMVA 2025**)
3. **Representation Learning Dynamics of Self-Supervised Models** Pascal Mattia Esser*, **Satyaki Mukherjee***, Debarghya Ghoshdastidar* TMLR 2024 Arxiv 2023
4. **Improved Representation Learning Through Tensorized Autoencoders** Pascal Mattia Esser*, **Satyaki Mukherjee***, Mahalakshmi Sabanayagam*, Debarghya Ghoshdastidar AISTATS 2023 Arxiv 2022.
5. **Point Spectrum of Periodic Operators on Universal Covering Trees** Jess Banks*, Jorge Garza-Vargas* **Satyaki Mukherjee*** IMRN 2021 Arxiv 2020.
6. **Gaussian Regularization of the Pseudospectrum and Davies' Conjecture** Jess Banks*, Archit Kulkarni*, **Satyaki Mukherjee***, Nikhil Srivastava*. CPAM 2021 Arxiv 2019.
7. **One-round discrete Voronoi game in $\mathbb{R}^2$ in presence of existing facilities.** Aritra Banik*, Bhaswar B. Bhattacharya*, Sandip Das* and **Satyaki Mukherjee*** In CCCG 2013, pages 37-42; CCCG 2013
8. **A note on the transversal size of a series of families constructed over Cycle Graph.** Kaushik Majumder*, **Satyaki Mukherjee*** JCTA 2018 ArXiv 2015

6 Academic Achievements and Awards

- Represented the Indian team and received Bronze medal at 51st International Mathematics Olympiad, Kazakhstan, 2010.
- Passed first round for the Informatics Olympiad selection procedure (organized by IARCS).
- Passed first round for the Physics Olympiad selection procedure (organized by IAPT).

7 Teaching Experience:

I was Graduate Student Instructor for the following courses.

- Math-54 : Undergraduate Linear Algebra
- Math-110: Upper division Linear Algebra.

I was an instructor at TUM for the following courses

- Praktikum for Analysis of new phenomena and reproducibility in AI/ML
- Seminar for theoretical advances in AI/ML

I have mentored various Bachelors, Masters and PhD students for reasearch Projects:

- Jing Bin Pan
- Gaurav Revankar
- Santanu Das
- Ramyak Bilas
- Hoang-Son Tran

8 Extra Curricular Activities:

- A principle organizer in the Singapore-Bangalore ”What is.. ?” workshop, a workshop geared towards fostering collaboration between NUS and research institutes in Bangalore.
- Was the student president in my undergraduate institution.
- Organized treasure hunts during my undergraduate studies.
- Was an organizer in the Mathematics Graduate Student Organisation, Berkeley.